

Day 1 Assignment

Complete Unit 1 – "Introduction to Agents & Vibe Coding"

- Listen to the [summary podcast](#) episode for this unit.
- To complement the podcast, read "[The New SDLC with Vibe Coding](#)" whitepaper.
- Complete these codelabs:
 - [Get started](#) with Antigravity 2.0 and IDE
 - [Build](#) a Web Application in AI Studio and Deploy to Cloud Run

What You'll Learn

Today's whitepaper talks about the transition from manual syntax coding to intent-driven "vibe coding" and disciplined "agentic engineering." It explores how AI agents compress the traditional software development life cycle (SDLC) and explains the "factory model" where developers act as system orchestrators who design the evaluation, constraint, and context harnesses that safely guide autonomous execution.

In the codelabs, you'll familiarize yourself with Antigravity 2.0, the Antigravity IDE, and the Antigravity CLI to vibecode your first applications. You'll also use Google AI Studio to deploy your app to Cloud Run, enabling you to share your vibe coded app with your friends.

Day 2 Assignment

Complete Unit 2 – "Agent Tools & Interoperability"

- Listen to the [summary podcast](#) episode for this unit.
- To complement the podcast, read the "[Agent Tools & Interoperability](#)" whitepaper.
- Complete these codelabs:
 - [Get started](#) with Antigravity CLI
 - [Explore](#) Google Developer Knowledge MCP server in Google Antigravity

What You'll Learn

Today's whitepaper talks about standardizing the plug-and-play AI ecosystem using open protocols to eliminate the complex technical debt of custom tool integrations. It details how the Model Context Protocol (MCP) connects models to data sources, outlines Agent2Agent (A2A) collaboration, showcases Agent-to-User Interface (A2UI) for generative UI, and introduces Agent Payments Protocol (AP2) and Universal Commerce Protocol (UCP) for secure machine-to-machine commerce.

In today's codelabs, you'll learn how to add MCP servers to Antigravity, giving it access to the canonical, machine-readable source of Google's public developer documentation. You'll then extend your familiarity with Antigravity by using it via the terminal with Antigravity CLI.



Day 3 Assignment

Complete Unit 3 - "Agent Skills":

- Listen to the [summary podcast](#) episode for this unit.
- To complement the podcast, read the "Agent Skills" whitepaper: <https://www.kaggle.com/whitepaper-agent-skills>
- Complete these codelabs:
 - [Explore](#) how Skills work in Antigravity
 - [Build](#) agents in Antigravity with Agents CLI and ADK



What You'll Learn

Today's whitepaper talks about managing dynamic context and avoiding "context rot" by equipping agents with portable "Agent Skills", directories structured around a central SKILL.md file. It explains how this framework uses progressive disclosure to keep system prompts lightweight, loading execution details and tools only on demand so that single agents can flex into hundreds of specialist roles efficiently.

In the codelabs, you will familiarize yourself with skills in Antigravity. Then, using Antigravity, you will install and use Agents CLI skills to create agents, lint your code, and test your agent, all by using natural language prompts.



Day 4 Assignment

Complete Unit 4 - "Vibe Coding Agent Security and Evaluation":

- Listen to the [summary podcast](#) episode for this unit.
- To complement the podcast, read the [Vibe Coding Agent Security and Evaluation](#) whitepaper.
- Complete these codelabs:
 - [Build](#) an expense-approval agent with human-in-the-loop triage and local evaluations with ADK, Agents CLI and Antigravity
 - [Write](#) Secure AI Code: Automated Threat Scans, Safety Guards, and Security Testing



What You'll Learn

Today's whitepaper talks about redefining security and evaluation in non-deterministic AI workflows by establishing continuous "Effective Trust" through a strict 7-pillar architecture. It outlines practical safeguards such as ephemeral sandboxing, defenses against hallucinated "slopsquatting" packages, an active Red/Blue/Green security triad, and OpenTelemetry trajectory evaluation.

In the codelabs, you will create an agent with human-in-the-loop so that humans can manually intervene, using ADK, Agents CLI and Antigravity. You will also learn about how to develop a secure, test-driven development with Antigravity.



Final Assignment

We've reached the final assignment! 🎉

Complete Unit 5 - "Spec-Driven Production Grade Development in the Age of Vibe Coding":

- Listen to the [summary podcast](#) episode for this unit.
- To complement the podcast, read the "[Spec-Driven Production Grade Development in the Age of Vibe Coding](#)" [whitepaper](#). Complete these codelabs:
- [Optional] [Deploy](#) and host your AI agents on Google Cloud
- [Optional] [Build](#) a front-end web app and link it to your cloud-hosted AI agent



What You'll Learn

Today's whitepaper covers bridging the gap between fragile vibe-coded prototypes and production-grade enterprise software using Spec-Driven Development (SDD). It details how to treat code as disposable and behavior-driven Gherkin specifications as the source of truth, establishing safe, zero-trust development pipelines with automated code-review agents and hybrid Policy Servers.

In the codelabs, you will learn how to create and deploy an agent to Google Cloud for enterprise scale. Then you will use Antigravity to vibe code a frontend client interface deployed to Cloud Run and tie it to an asynchronous event-triggering architecture that automatically feeds live expense submissions straight to your cloud-hosted agent. While these codelabs are optional today because they may require a billing account on Google Cloud, we still encourage you to scan through them to familiarize yourself with what productionization of agents and apps might look like at enterprise scale.